Emery Dittmer
260658030
MGSC 670- Revenue Management

Revenue Management

Individual Assignment 3 – Revenue Management
Based on the article "Frustration-Based Promotions: Field Experiments in Ride-Sharing"

1 What is the motivation of this paper? Please elaborate.

This paper interrogates the possibility that companies address customer satisfaction issue proactively.

Customer satisfaction is a very difficult metric to measure accurately and fully understand. Customers who file complaints are heard; however, when they do not voice their complaints, there is no record of frustration with the service or product. Unhappy customers are likely to decide to stop using the service.

Companies need help addressing customer satisfaction, especially when they do not necessarily know who is unhappy or happy. The paper studies a company *Via* which is a ride-sharing platform. Within the transportation industry, ride-sharing has become popular with companies like Uber or Lyft. A subsegment of ridesharing called carpooling is where Via operates. Carpooling lets several users share the same trip to and from various locations or along the same route.

However, the experience can be challenging. Via and the researchers identified several pain points, such as long wait times, many stops, significant detours, and no drivers showing up. They focused on long wait times and long travel times. To better define the long wait times, the researchers used a metric called the estimated time of arrival error or ETA error, which predicts the difference between the estimated ETA communicated to the user and the actual arrival of the vehicle. Next, the research defines Via-Google Ratio or VGR, with computes the difference between a Via trip and an equivalent trip based on Google Maps under the same conditions. If a Via trip took twice as long as a google maps route, then the VGR is 2. VGR is a stand-in to capture the number of detours and time for loading and unloading passengers who are in the current customer's journey.

The study examined customers whose VGR was above 2 or whose ETA error was above 10 minutes. These customers were then randomly assigned to one of 4 groups. 3 of these groups received a treatment and one was left as a control from comparison. The study determines the impact of a proactive approach by either communication to the customer, a waived ride fee, a credit to a future ride or no action.

2 What are the differences between Experiments 1 and 2?

There are several key differences between experiments 1 and 2.

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First, both experiments are in different geographies with different city demographic features. Experiment 1 is based in New York City (NYC), while Experiment 2 is based in Washington DC (DC). The density of NYC is much higher than that of DC. Additionally, the local population of the cities has a different relationship with transportation, as in NYC, most trips tend to be quite short, less than 12 minutes.[1]

Second, there are differences in the impact of the user base. DC has a relatively low Via maturity, while NYC has a much higher level. This means there are fewer rides per capita, and total in DC than in NYC. This difference may be crucial when balancing the randomization of users and considering the experimental setup.

Lastly, in terms of the experiment, the NYC user group had a control with 3 treatment options, DC added a treatment option. The NYC treatment groups were randomly given a communication, a waived ride fee or a credit to future purchases. The DC treatment groups, meanwhile, were given an additional grouping of a discount on the next ride. A discount of 50% would be applied for the next ride.

3 What is the goal of conducting balancedness tests in a field experiment?

The goal of conducting balancedness tests is to showcase the validity of the randomization of the users into separate groups. Users must be randomly assigned to groups to ensure meaningful insights from the data. By randomizing treatments and controls the data for each treatment and control group will be “balanced”. In other words, the data will have less bias since each user group or treatment will be averaged out and contain equal types of uses or equal bias. Remove any bias between groups and get statistically significant results. Without knowing that the data is balanced, the results will be skewed, and so will the insights. The classic data saying “garbage in, garbage out” applies here. If the data used for insights is not balanced, biasing or other factors may be underlying the data. Any analysis will Therefore, it Is crucial to do A/A testing or ensure that the data entering the

For instance, within this study, a mishap with the VGR group led to some customers receiving the credit condition versus the waived condition. For this situation robustness test were performed to ensure that the group of users within this group of credit treatment was statistically similar to other groups and that there was no “contamination” of data.

4 According to the results of this paper “The best intervention is to offer a credit for future usage.” What are the possible explications behind this result? Do they match your expectations/intuitions? Please elaborate.

There are three key explanations behind this result. The rarity of service gives credibility to the apology, the future-looking nature of credits and the endowment effect.

First, service delays and interruptions are relatively rare. The rarity of service interruptions means that an apology will be effective as the service disruption is the exception rather than the

rule. In settings with high levels of service disruption, apologies negatively impact customers as per the literature.[2,3]

Second, service users may have a better psychological interpretation of the apology with the credit as opposed to any other treatment. The communications and the waived fees are rear-looking adjustments to the service quality level, while the credit is forward-looking. The credit and message psychologically price customers to look forward or apply their discount on future rides. In other words, the treatments reflect different virtue signalling from *via* to customer, impacting their psyche. The communications and waived fee strategies communicate to the customer that the previous negative experience is the exception to the rule and that via hold themselves to a high standard of no errors. This message comes from the idea that via is apologizing for something in the past. The credit communicates indirectly that the service was not to the company's standards, but via is so confident they will pay for the next trip. The credit, therefore, signals moral correctness to make up for the service while anthropomorphizing *via .* In theory, the indirect message is "mistakes happen, but we are confident this is not normal. We are so confident we are willing to take care of your net ride".

Lastly, credit for a ride creates a feeling of obligation within the consumer. The endowment effect is theorized to be a loss aversion technique whereby people value objects they own at a higher rate than the same objects they do not own.[4] In this case, customers likely place a higher importance on the credit that belongs to them, then the waived ride that cannot be owned. The ownership of an object or idea is a psychological motivation deeply embedded in humans that may link to increased ridership and customer revenue after the credit treatment.

These treatments do not match my intuition. The likelihood of increased ridership and future uses is proportional to the monetary value of the treatment. In other words, the $5 credit would have an effect of 5% retention, while a waived car ride would depend on the cost of the car ride. I am someone who takes careful consideration of my ownership and finances. *Via* operates on a transactional nature; therefore, I expect its consumers to use and think of the service in much the same way; transactional. Therefore, a likelihood of future ridership correlated to absolute monetary value would make sense.

5 Why do you think that the intervention was successful for the ETA-error frustration but not for the VGR frustration?

A high VGR likely indicates many stops on the route for the customer. I exepct the difference here lies in the perceived intention and control of the error. ETA error is not necessarily in *Via*'s control. The control lies in exogenous factors such as traffic, the driver, or other roadside events. ETA is an estimate that may be highly acuracte but inherently unpredictable by via or current means.Therefore, customers will likely internalize or understand to some degree that an ETA error is an unintentional goof.

However, VGR is under the influence of *Via.* Let's assume that the VGR is below 1.5 on a trip that does not pick up additional passengers. Now let's assume that VGRs above 1.5 are directly proportional to the number of additional passengers and the deviation from the ideal route for passengers within the ride. The route and corresponding VGR are directly in the

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260658030
MGSC 670- Revenue Management

control of *Via* and their technology. Therefore, a high VGR is strongly influenced by *Via* and the decisions to pick up additional passengers. The intention of *Via* from the consumer's perspective changes from providing a service for them to providing a service and optimizing for profit. Regardless of treatment, the ensuing apology is given with the intentional context, which may lead the consumer to blame Via instead of adopting an understanding. In other words, the reaction from the consumer may be to question why this inconvenience was not simply avoided, given that the company is aware that this frustration comes from something within VGR's control.

6 What can be the long-term effects of the interventions used in this type of field experiments?

Via and others may adopt the dissatisfaction anticipation approach and offer credits to consumers with high ETA error 1 hour after the incident. Companies adapting this approach to their industry will likely see that this will increase profit while decreasing minor service-related complaints.

However, this will also give rise to the anchoring effect whereby consumers expect a discount or credit when their ETA error or equivalent metric is high. There are also ethical consumers around treating different user segments (high usage, medium usage, low or new usage) differently, as users will likely be frustrated with the lack of consistency based on their usage. This complex decision-making policy will likely end with users with a longer tenure with e app, which may demand more regard while new users may see little impact. Regardless, the practice is morally questionable.

In the long term, these interventions will benefit the company by heling the ridership , profitability. However, there may be unintended consequences that would require careful business practice and rule consideration.

7 Provide three examples of industries (or services) where frustration-based promotions can be an effective strategy. Please provide a detailed answer (potentially with supporting evidence from the Internet).

7.1 Airlines

Some airlines offer proactive compensation for cancelled flights. Customer service within the airline industry is robust but deals with a lumpy supply problem, where problems often affect all passengers on a flight. Therefore between 1 and a couple of hundred of people are usually affected by issues meaning that many agents are needed for a short time and then not the rest of the time. To approach this, Delta offers bonus miles for flights It has cancelled. While this may not address the rebooking issue, this does give a credit to the consumers and makes the interaction with customer service easier, and a decision to fly Delta again much easier.[5]

7.2 Telecommunications

Some telecommunication companies will offer credits or discounts for large-scale outages that affect many people and could have been avoided. For instance, in July 2022 a

Emery Dittmer
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MGSC 670- Revenue Management

massive outage for Rodgers in Toronto, Canada, caused millions of customers to lose cellular and internet connectivity from Friday to Sunday. This disrupted productivity and livelihood throughout the region. The company offered a credit of 5 days as an apology and to make up for the service disruption. [6]

7.3 Amazon

Online retailer Amazon has a firm policy of customer service. The proactive approach is made to save on costs and keep customers happy. Amazon has 6 customer service tents that dictate how to understand and treat customers. One tangible outcome of this according to Forbes is their refund policy for movie downloads when the download is slow. [7] Amazon has also been known to alert customers, offer free amazon prime (a premium subscription service) or credits proactively when a package is late.[7, 8]



Figure 1: Amazon Customer Tenants.

Customer tenants reproduced for legibility:

- "Relentlessly advocate for customers."
- "Trust our customers and rely on associates to use good judgement."
- "Anticipate customer needs and treat their time and attention as sacred."
- "Deliver personalized, peculiar experiences that customers love."
- "Make it simple to detect and systematically escalate problems."
- "Eliminate customer effort through this sequential and systematic approach: defect elimination, self-service, automation, and support from an expert associate."

8 Focus on one example from the previous question and design a field experiment where you define the frustration types, the relevant interventions, and the appropriate engagement metric(s)

We will be using the example of Amazon for this experimental design. Amazon has grown from a book delivery service to much more. They offer product delivery, movie downloads, music, books through their kindle application, apps and games store. For this experiment, we will focus on the book and app stores. These two items are conceptually the same. They are both consumable online content for the user. In this context, the biggest frustration with online content purchases, be it apps games or books, is that the content does not meet the user's expectations. When the user is frustrated, they will generally

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MGSC 670- Revenue Management

stop engaging with that content. Many times, users are encouraged to write reviews, but few do. In fact, on average, between 5-10% of users will end up writing product reviews. [9] The lack of reviews demonstrates the silent frustrated consumer, but also damages any potential improvements on content that could be made.

Often, the frustration with the product is felt from the customer as expectations letdown or lack of engagement with the material and at times frustration. The most applicable for an expiramental setup is user engagement. Therefore, we will examine the frustration of a customer by the total user engagement in the number of hours post-buying and time since last engagement. The underlying idea is that if the engagement is less than 2 hours and the last engagement date was 7 days, then the customer is frustrated and the material is forgotten. The 7-day cycle represents the time in a week and is designed to accommodate various lifestyles, including those with demanding work schedules.

To best capture the data, we will capture data for users and find users who have experienced a 'frustration' with a product where they have not engaged for more than 2 hours with a purchased app or book and have left it untouched or unopened for more than 7 days. These users will randomly be split into one of the following groups. Users will need to be carefully selected as those who have not had frustration previously, are able to be tracked for the duration of the experiment and other factors. Balancedness testing and other statistical measures will need to be conducted to validate statistical similarly between groups.

The users will be randomly assigned a group. There are 5 potential groups a user will fall into. A control group where nothing is done, a notification group who will receive a notification that their recent purchase has been unattended, a refund group who will be offered a refund for the product they have bought (this group will be made up of users who follow through with the refund and those who do not) and a 10% discount group and lastly a group who will receive an offer to write a review for 5% off their next purchase.

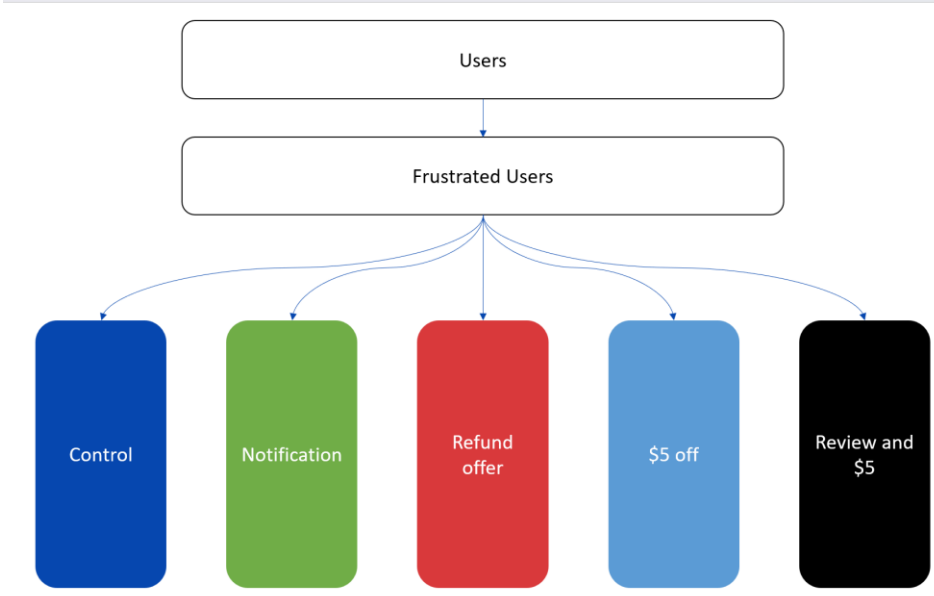


Figure 2: Basic Experimental design

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The idea behind each treatment is as follows:

The control is a baseline for users.

Notifications are a “cheap” solution to let the user know their content is unused to see their reaction.

The refund offer allows the user to choose whether or not to receive a refund and is an approach some companies like steam , an online video game retailing platform, offer. The idea is a trial period. If you do not like the product within 4 hours of playing then you can return it.

The $5 off policy simulates the *Via* case study policy, although it may bring up significant concerns about anchoring or expectations for consumers.

The last policy, the review and $5 credit is designed to encourage continual content creation. Amazon and other app marketplaces profit from volume and number of apps sold. Therefore, if amazon can help encourage the conversion of users to write reviews through incentives, the app creators may have additional content and feedback to improve. This business model is much longer term and if productionized would need to be limited in the number of users or risk anchoring effects. This strategy could ultimately prove more beneficial to their business strategy if well executed.

The main metrics to investigate for this study are the effects on future purchases in dollar amounts and the quantity of downloaded content. These key metrics should both increase higher for a treatment option than for the control over the following 1-52 weeks. The number of apps downloaded would however need to be categorized into free and paid apps or books. Additional complications may arise with the freemium model where in-app purchases are made. The in-app purchases can be tracked separately, eliminated or added to the total money paid. Additionally, considerations would need to be made for the number of reviews and likelihoods of writing reviews (defined as the number of reviews % quantity of material downloaded).

Overall this model of proactive customer satisfaction is easy to deploy as all purchases are electronic and is feasible with the technology and current data available. The assumption of frustration may benefit from user surveys and focus groups. However, the 5 treatment options represent different approaches and should instigate different responses from the consumer vis a vis their frustration. An effective strategy may also help amazon grow their app marketplace traffic away from the Google and Apple, the two dominant players/

9 References

(1) Sibal, A. NYC Taxi Trip Duration Prediction using Machine Learning. *Medium* **2022**.
(2) Schweitzer, M. E.; Hershey, J. C.; Bradlow, E. T. Promises and lies: Restoring violated trust. *Organizational behavior and human decision processes* **2006**, *101* (1), 1-19.
(3) Ho, B. Apologies as signals: with evidence from a trust game. *Management Science* **2012**, *58* (1), 141-158.
(4) Van Dijk, E.; Van Knippenberg, D. Buying and selling exchange goods: Loss aversion and the endowment effect. *Journal of Economic Psychology* **1996**, *17* (4), 517-524.

Emery Dittmer
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MGSC 670- Revenue Management

(5) Schlappig, B. Delta's Proactive Flight Cancelation Compensation. *One Mile at a time*, 2023.
(6) Gollom, M. Rogers to credit customers 5 days of service for outage. *CBC News*, 2022.
(7) Gingiss, D. 6 Customer Service Tenets Used By Amazon To Create Effortless Experiences. *Forbes*, 2019.
(8) Roosa, A. Proactive customer service: Definition, strategies, and examples. Zendesk Blog, 2022.
(9) Hester, P. 26 Mind-Boggling Online Review Statistics & Facts for 2023. wEBSITEBUILDER, 2021.

Link for ref 5

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